Block practice · Integration, areas, volumes and kinematics

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

Never at rest, and the least speed

OLevels.com SCHOOL

Practice workbook · Never at rest, and the least speed

Using the discriminant, completing the square or an inequality on v or a to describe the motion.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

Type ID: FPM-CAL-27

1. Question 7 · part d.ii · 3 marks

2023 JAN · Paper 01 (regional R)

Worked-example retry · Attempt part d.ii only.

7 A particle P is moving along the x-axis. At time t seconds $(t \geqslant 0)$ the velocity of P is v m/s where

$$v = t^2 - 10t + 28$$

(a) Find the velocity of P when $t = 1$

(1)

Given that the distance of P from the origin is 24 m when $t = 3$

(b) find the distance of P from the origin when $t = 5$

(5)

(c) Find the acceleration of P when $t = 9$

(2)

(d) (i) Show that there are no values of t for which P is instantaneously at rest.

(ii) Find the least magnitude of the velocity of P

(3)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

1. Question 7 · part d.ii · 3 marks

2023 JAN · Paper 01 (regional R)

Worked-example retry · Attempt part d.ii only.

Question 7 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

1. Question 7 · part d.ii · 3 marks

2023 JAN · Paper 01 (regional R)

Worked-example retry · Attempt part d.ii only.

Question 7 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

16

P 7 1 8 1 8 A 0 1 6 2 8

1. Question 7 · part d.ii · 3 marks

2023 JAN · Paper 01 (regional R)

Worked-example retry · Attempt part d.ii only.



1. Question 7 · part d.ii · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 1 · part a · 2 marks

2025 JUN · Paper 01

Further practice · Attempt part a only.

1 A particle P moves in a straight line. At time t seconds, the velocity, v m/s, of P is given by

$$v = 3t^2 - 9t + 7$$

(a) Show that P never comes to rest.

(2)

(b) Find the acceleration of P, in m/s^2, when $t = 4$

(2)

Using algebra

(c) find the distance, in m, that P travels in the interval $0 \leqslant t \leqslant 5$

(3)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

2. Question 1 · part a · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 4 · part a · 3 marks

2025 NOV · Paper 02

Further practice · Attempt part a only.

4 A particle P is moving along a straight line. At time t seconds $(t \geqslant 0)$, its velocity, v m/s, is given by $v = 4t^2 - 6t + 5$

(a) Find the minimum speed of P

(3)

The acceleration of P at time T seconds is $18\ \text{m/s}^2$

(b) Find the value of T

(2)

When $t = 0$, P is at the point X, and when $t = 3$, P is at the point Y

(c) Find the distance XY

(4)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

3. Question 4 · part a · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 7 · part d.i · 0 marks

2023 JAN · Paper 01 (regional R)

Further practice · Attempt part d.i only.

7 A particle P is moving along the x-axis. At time t seconds $(t \geqslant 0)$ the velocity of P is v m/s where

$$v = t^2 - 10t + 28$$

(a) Find the velocity of P when $t = 1$

(1)

Given that the distance of P from the origin is 24 m when $t = 3$

(b) find the distance of P from the origin when $t = 5$

(5)

(c) Find the acceleration of P when $t = 9$

(2)

(d) (i) Show that there are no values of t for which P is instantaneously at rest.

(ii) Find the least magnitude of the velocity of P

(3)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

4. Question 7 · part d.i · 0 marks · Working and answer

Working and answer

OLevels.com SCHOOL